The Preventable Chargeback: A Prevention Roadmap

Chargeback Prediction Model — Cinderhaven Data Platform

Prevention Roadmap

$446,200 in chargebacks over 37 months. 73% is preventable.

Cinderhaven carries **$446,200** in retailer chargebacks over 37 months (Jan 2023 – Jan 2026) — an annualized $144,714 a year. Grouping the chargebacks by root cause and applying a fixed preventability fraction to each category identifies **$324,888** — 73% of the total — as addressable through four fixable upstream data conditions. This figure is an assumption-driven estimate (historical loss per archetype × its preventability fraction), not a model prediction.

The scoring workflow names the specific data condition driving each purchase order's risk. This report presents the root-cause breakdown, the ranked prevention roadmap, and the economics of fixing versus disputing.

This analysis uses Cinderhaven synthetic demonstration data to illustrate a repeatable methodology. The numbers, proportions, and root-cause distribution reflect a realistic CPG brand profile.

Root-cause breakdown

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(a) Historical chargeback amount by root cause. Logistics overage — primarily missing case dimensions and case weight — accounts for the single largest preventable category.

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(b)

Figure 1

Margin math

The table below translates each root cause into a dollar figure: what it cost historically, what fraction is preventable with targeted data fixes, and the estimated recovery over the 37-month window.

Table 1

Table 1

Root cause	Historical loss	Preventable	Prevention value	Chargeback count
Logistics Overage	$245K	70%	$171K	2284
Data Compliance Error	$143K	80%	$114K	300
ASN Timing Infraction	$42K	70%	$29K	273
Pricing Discrepancy	$17K	60%	$10K	16

Ranked prevention roadmap

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Figure 2: Prevention value by root cause — estimated 37-month dollar recovery from targeted data fixes. Logistics overage tops the list at $171K in prevention value ($245K historical loss), driven primarily by case dimension and case weight gaps in the product master.

The actions required to capture this prevention value are data management tasks, not operational overhauls. The roadmap below names the specific fix for each root cause.

Table 2

Table 2

Rank	Root cause	Prevention value (37-mo)	Required action
1	Logistics Overage	$171K	Audit pick-and-pack procedures; add case-count verific
2	Data Compliance Error	$114K	Populate missing barcodes (GTIN-14, UPC) and verify
3	ASN Timing Infraction	$29K	Submit Advance Ship Notices within the retailer-requi
4	Pricing Discrepancy	$10K	Reconcile PO unit prices against approved price lists b

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Recovery versus prevention

The deductions team currently disputes chargebacks after they post. The industry win rate on retailer disputes is 30–45%. At a 40% win rate, recovering the $446,200 in chargebacks accrued over the 37-month window yields roughly **$178,480** — before dispute labor costs.

Prevention requires no dispute process. Fixing the data conditions identified above eliminates the chargeback at source.

A companion model demonstrates pre-ship scoring, with one caveat stated plainly. On Cinderhaven's *real* chargebacks the features carry no predictive signal (AUC 0.50): real chargebacks are effectively random with respect to data quality. The model is therefore trained on **synthetic** labels drawn from a known risk formula (risk ASN lateness × data-quality gaps), and it recovers that formula. On held-out synthetic data its AUC is **0.70** — a *ranking* measure: it ranks a charged-back shipment above a clean one 70% of the time. At a 0.5 threshold it flags 72% of chargebacks (recall). AUC is not the flag rate, and predictive skill on the real target is effectively zero — the durable value is the prevention roadmap, not the score.

The math is straightforward: **$324,888** in preventable chargebacks recovered at 100% is worth more than $446,200 recovered at 40%, and it eliminates the working capital drag of post-ship disputes.

This analysis was produced using Cinderhaven synthetic demonstration data. All dollar figures, chargeback counts, and root-cause distributions are illustrative of a real methodology, not actual Cinderhaven financials. Real chargebacks carry no predictive signal (AUC 0.50); the model is trained on synthetic labels and the reported AUC is measured on held-out synthetic data.